Data on Earthquakes

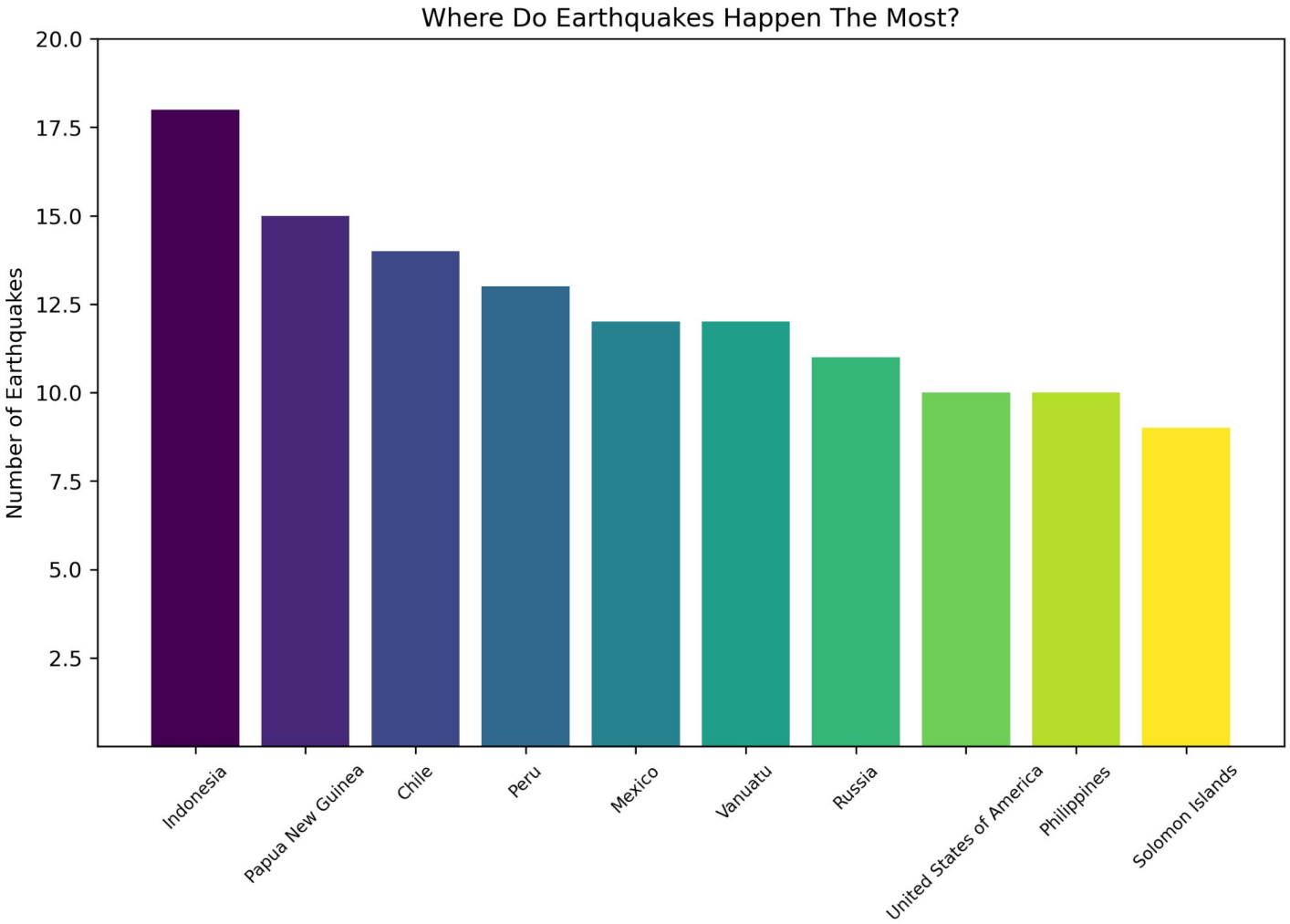
The Data

This presentation’s purpose is to showcase findings of analyzed Earthquake data. The data spans from years 1995 - 2023.

- This dataset contains over 1000 Earthquakes documented.
- All continents are included for a true global analysis.
- Everywhere from magnitude to magnitude type and alert levels are included in the recordings.

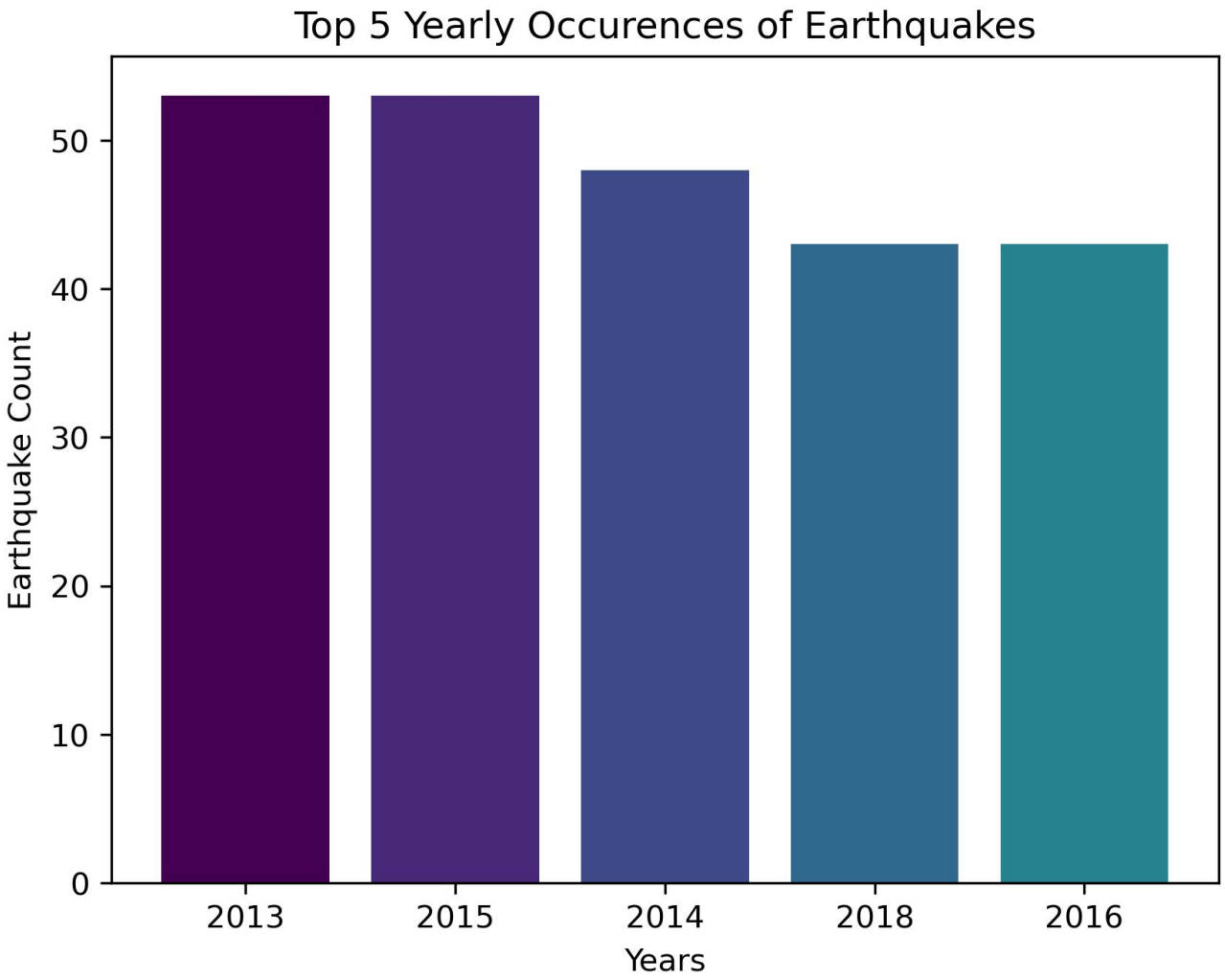
Where Do Earthquakes Happen The Most?

For this presentation expect to see a quite basic overview of the data. There was no need for specifics or to over-complicate the findings. Checkout the first graph of findings on the next slide!


Where Do Earthquakes Happen The Most?
Number of Earthquakes
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Indonesia
Papua New Guinea
Chile
Peru
Mexico
Vanuatu
Russia
United States of America
Philippines
Solomon Islands

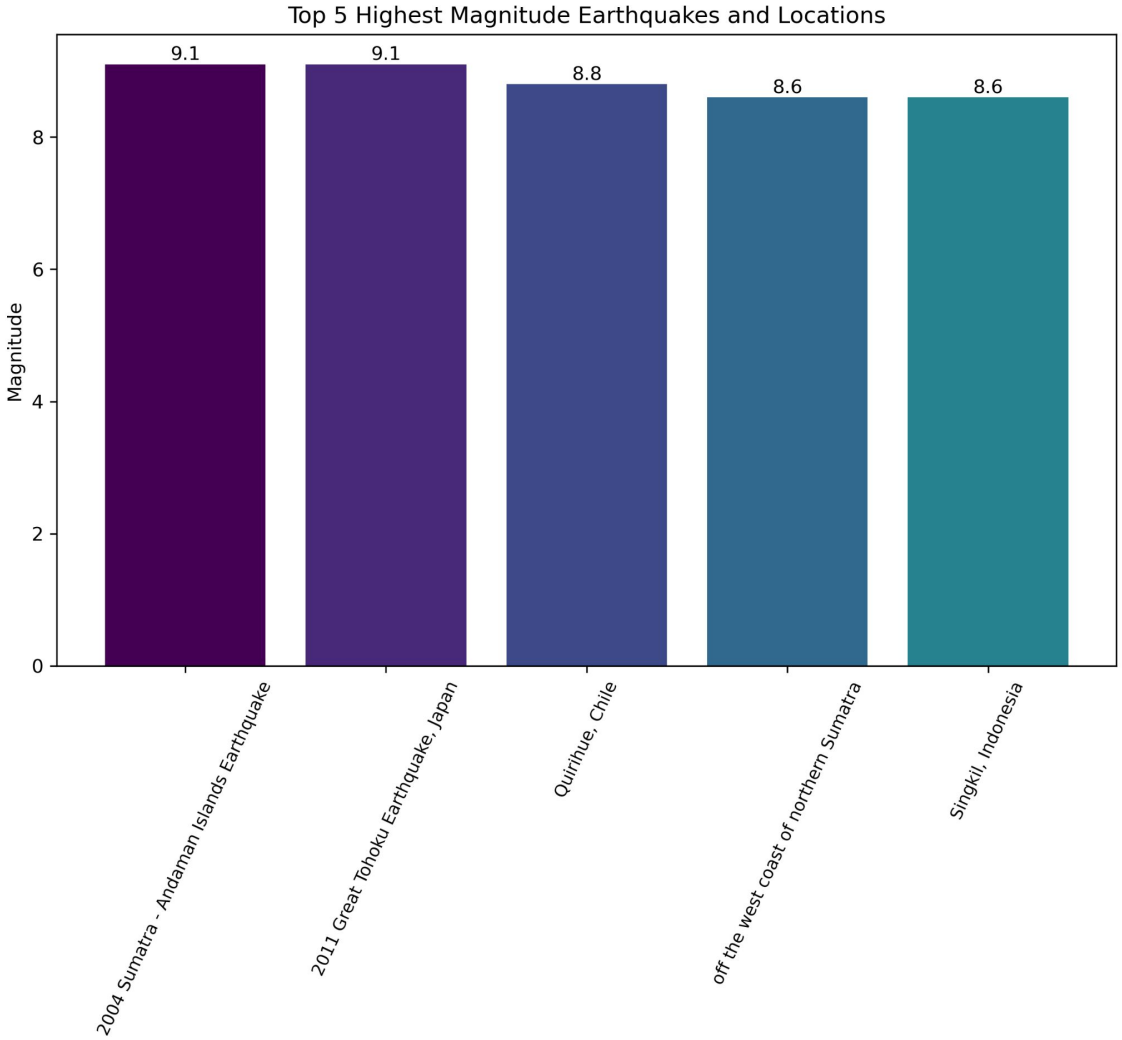
What Years Do Earthquakes Happen The Most?

Knowing where all of the Earthquakes tend to happen. The next question to follow that is when? Kindly skip to the next slide to see the data:


Top 5 Yearly Occurences of Earthquakes
Earthquake Count
0
10
20
30
40
50
2013
2015
2014
2018
2016
Years

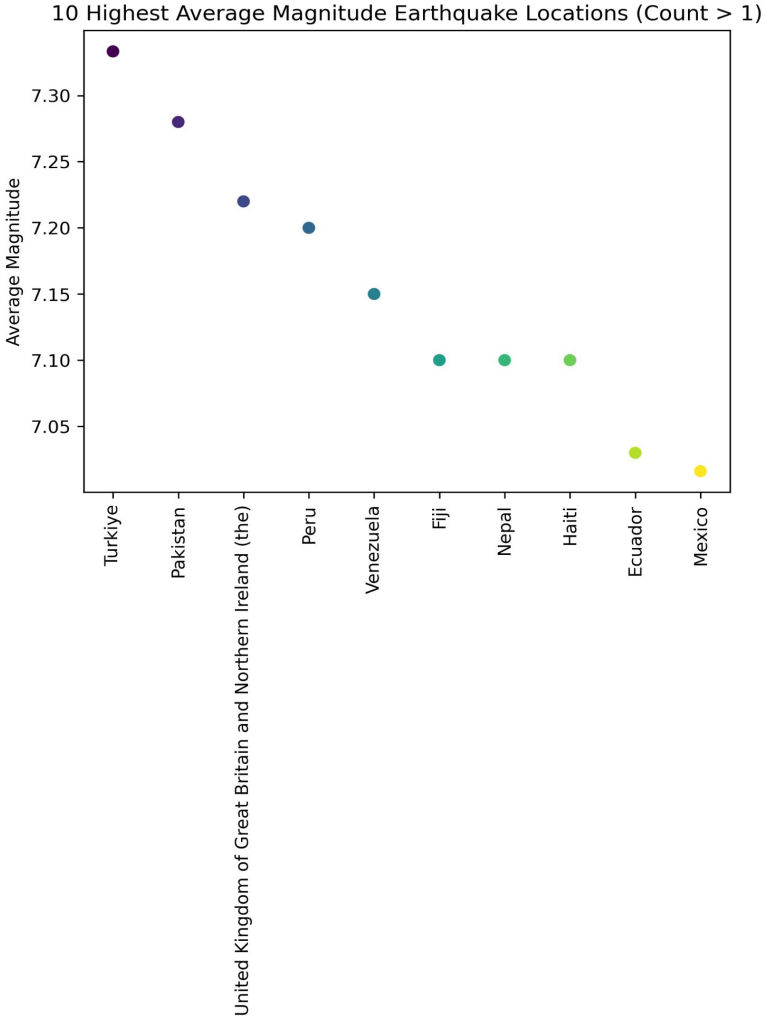
What Is Considered Most Dangerous?

After figuring out where Earthquakes highly occur and the years when Earthquakes happen the most, the next question to ask is where are the highest magnitude Earthquakes?


Top 5 Highest Magnitude Earthquakes and Locations
Magnitude
0
2
4
6
8
9.1
9.1
8.8
8.6
8.6
2004 Sumatra - Andaman Islands Earthquake
2011 Great Tohoku Earthquake, Japan
Quirihue, Chile
off the west coast of northern Sumatra
Singkil, Indonesia

Last Question

The final question that was drawn is ‘where the highest magnitude earthquakes happen on average?’. Knowing what we know about the highest magnitude earthquakes, do locations correlate or do earthquakes with high magnitudes happen at random?


10 Highest Average Magnitude Earthquake Locations (Count > 1)
Average Magnitude
7.30
7.25
7.20
7.15
7.10
7.05
Turkiye
Pakistan
United Kingdom of Great Britain and Northern Ireland (the)
Peru
Venezuela
Fiji
Nepal
Haiti
Ecuador
Mexico

Conclusion

The data shows that overall earthquakes do not happen in a predictable manner. From where the most earthquakes happen to the highest magnitudes, where the most prominent data metrics show up are always changing.

So, earthquakes truly are a freak of nature. Though with more questions and data we can hope to learn more about their nature.

Thank You For Your Time!

Sincerely,

CJ